

Modified Truss & CHB Gable Structure - Project: bar_5m

1. Global Building Parameters

- Building Span:** 20.0 m
- Total Bays:** 4 bays @ 5.0 m spacing (Total Length: 20.0m)
- Main Steel Frames:** 3 Interior Trusses (5.0m Tributary Width)
- Gable Ends:** 2 Reinforced CHB Walls (Bearing 2.5m Tributary Width)
- Roof Pitch:** 10.0 degrees
- Truss Depth (Parallel):** 0.80 m
- Apex Height (Total):** 6.96 m
- Column/Eave Height:** 5.20 m
- Number of Panels:** 14 (Per Top/Bottom Chord)

2. Interior Roof Truss Member Selection (AISC Chapter H)

- Max Deflection:** 0.13 mm

Group	Selected Shape	Weight (plf)	Area (in²)	Interaction Ratio
Chords	WT4X14	14.0	4.12	0.98
Webs	2L2X2X3/16	4.9	1.44	0.95

- Total Building Steel Weight (3 Trusses):** 3,287.4 kg
- Estimated Steel Material Cost:** PHP 213,677.92 (@ PHP 65/kg) (Note: Cost excludes the front & rear concrete gable walls)

3. Global Steel Truss Cutting List (3 Trusses)

Component	Shape	L per Piece (m)	Qty per Truss	Total Building Qty	Total Length (m)
Top/Bot Chords	WT4X14	1.451	28	84	121.85
Vertical Webs	2L2X2X3/16	0.812	15	45	36.56
Diagonal Webs	2L2X2X3/16	1.643	14	42	69.02

4. Front & Rear CHB Gable Wall Specification

The 2 end steel trusses have been omitted in favor of structural Concrete Hollow Block (CHB) walls that directly carry the roof purlins and resist lateral wind loads across the 20m span.

RC Column Layout (Vertical Supports)

- Quantity:** 6 structural concrete columns per gable wall.
- Spacing:** Maximum 4.0 m horizontal spacing between columns.

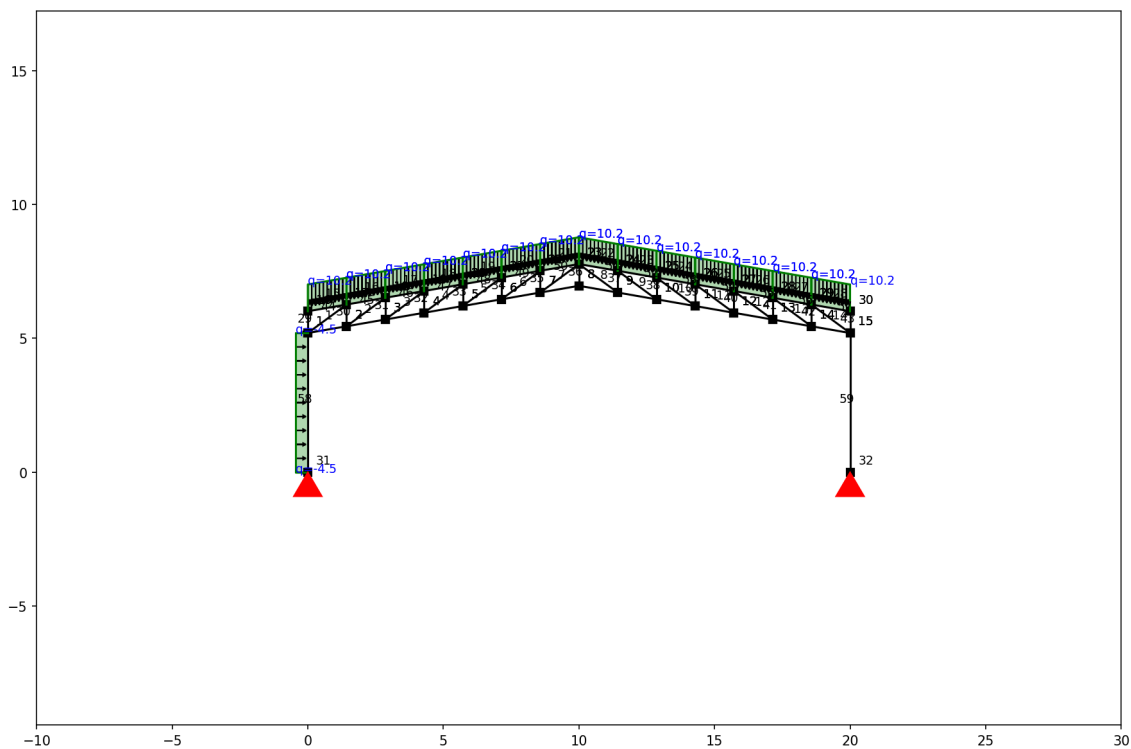
- **Column Sizing:** All gable wall columns to be 200mm x 350mm reinforced with 8 pcs 16mm ϕ vertical rebars.

RC Beam Layout (Horizontal Supports)

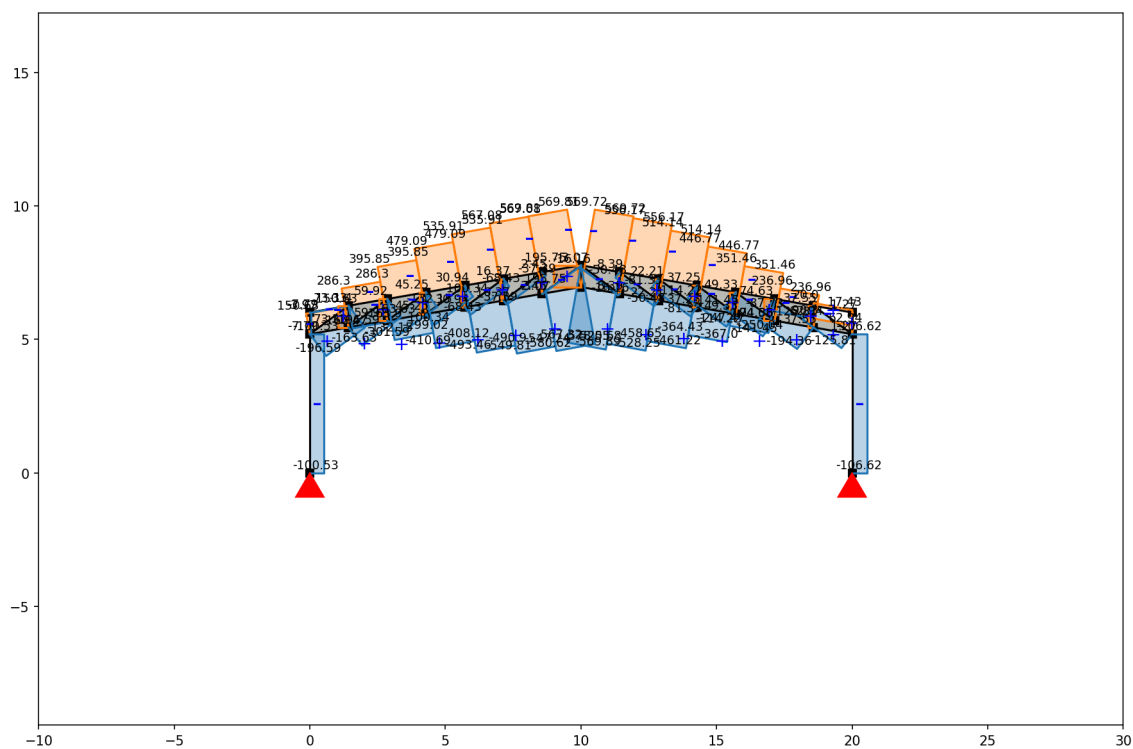
- **Mid-Height Tie Beam:** Continuous 150mm x 300mm beam at approx 2.8m - 3.0m height to break the unbraced CHB vertical span.
- **Sloped Gable Roof Beam:** Continuous 200mm x 400mm boundary beam tracing the 10-degree roof slope (from 5.2m eave up to 6.96m apex).
 - **Main Reinforcement:** 6 pcs 16mm ϕ continuous long bars (3 top, 3 bottom).
 - **Stirrups:** 10mm ϕ ties spaced at 100mm near columns and 200mm at midspan.
 - **Embedment:** Purlin cleats/plates must be embedded here at 0.60m max spacing.

5. Visualizations (Interior Truss)

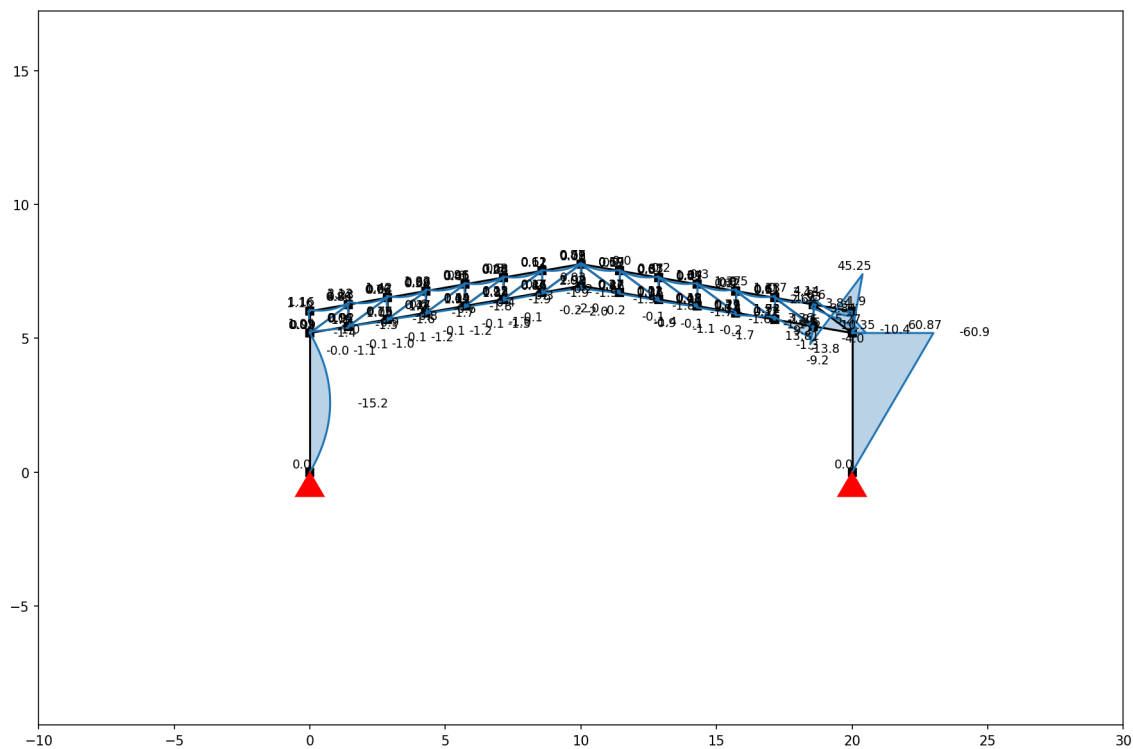
Truss Structure & Loads



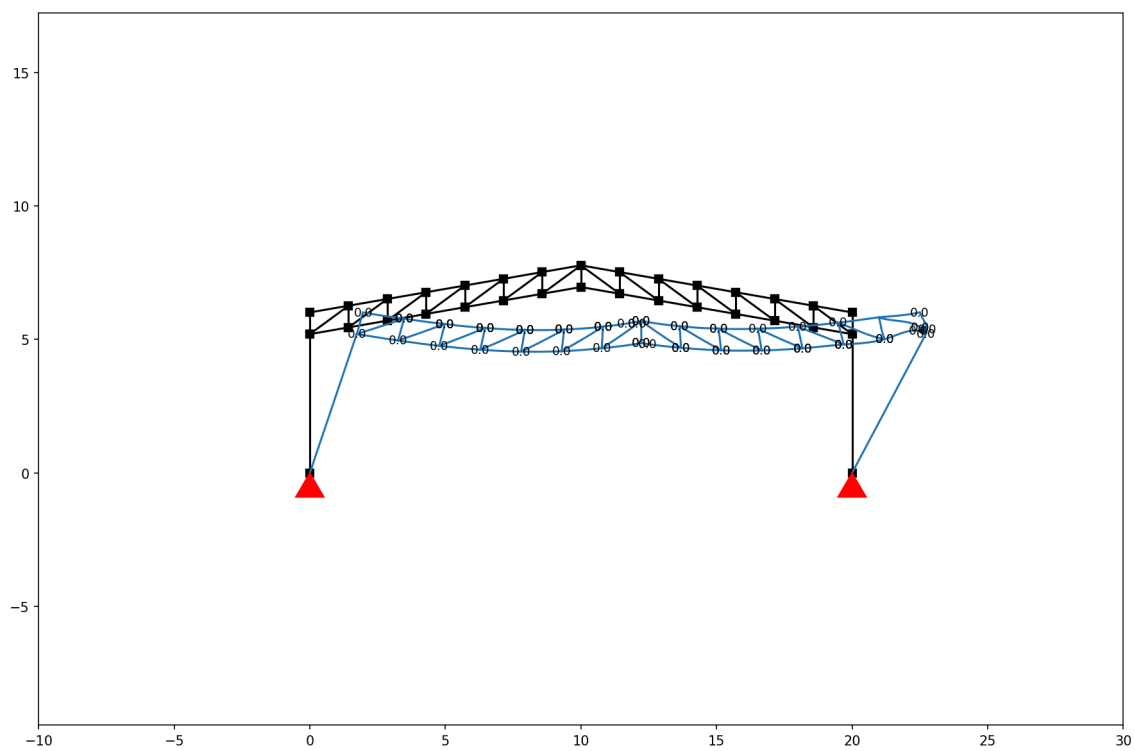
Axial Force Diagram



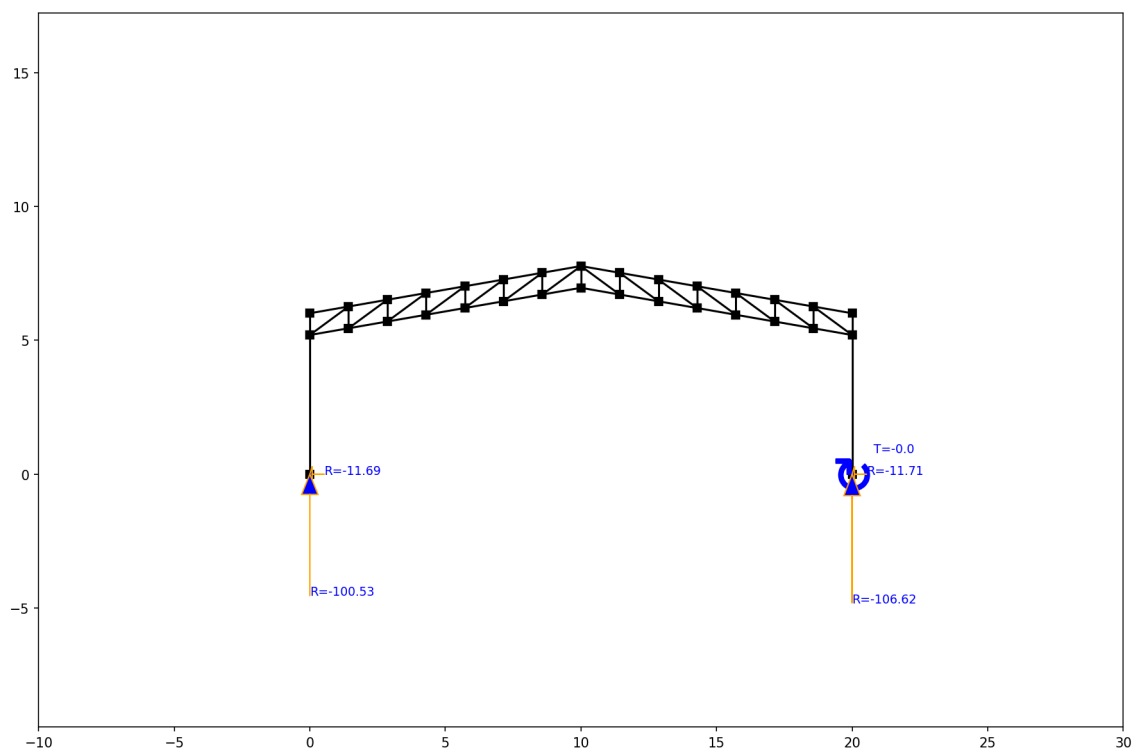
Bending Moment Diagram



Displacement Plot



Reaction Forces



6. 3D Bracing Requirements (3-Truss System)

To ensure global stability out-of-plane for the 4-bay building, the following bracing is required:

- **Roof Purlins (Top Chord Bracing):** LC 150x50x20x1.5mm spaced at max 0.60m attached to truss chords and the CHB Gable Beam.
- **Bottom Chord Kickers (Uplift Bracing):** L2X2X1/4 angle kickers installed every 3.0m along the internal truss bottom chords.
- **Longitudinal Roof X-Bracing:** 16mm ϕ sag rods spanning across the roof plane in the 1st and 4th bays (adjacent to the stiff CHB gable walls).
- **Sidewall Vertical X-Bracing:** 20mm ϕ rods or L2X2X1/4 angles in the 1st and 4th wall bays to transfer longitudinal shear.
- **Eave Struts:** Continuous longitudinal members connecting the RC columns to the rigid CHB corner columns.